

CONCEPT QUESTIONS – CH : 2

Introduction

2-1C

How does transient heat transfer differ from steady heat transfer? How does one-dimensional heat transfer differ from two-dimensional heat transfer?

Transient occurs when there is a temperature change over time. Steady state occurs when temperature is constant. Heat transfer in one-dimensional occurs in one direction while in two-dimensional occurs in two directions with relative magnitudes.

2-2C

Is heat transfer a scalar or vector quantity? Explain. Answer the same question for temperature.

Heat transfer is a vector since it has a magnitude and direction. Temperature is a scalar since it only has a magnitude.

2-3C

Does a heat flux vector at a point P on an isothermal surface of a medium have to be perpendicular to the surface at that point? Explain.

Yes, a heat flux vector must be perpendicular to the surface. Since the heat flux is proportional to the temperature gradient it points in the direction of highest rate of accent which would be perpendicular.

2-4C

From a heat transfer point of view, what is the difference between isotropic and anisotropic materials?

Isotropic have same properties in all directions. An Anisotropic material the properties change with direction. For example, thermal conductivity will remain constant for isotropic and have variation in anisotropic.

2-5C

What is heat generation in a solid? Give examples.

A medium which other form of energy are generated to create thermal energy within the solid. Some examples include electrical resistance, chemical reactions, and nuclear reactions.

2-6C

Heat generation is also referred to as energy generation or thermal energy generation. What do you think of these phrases?

The terms energy generation & thermal generation implies that energy is being created whereas in heat generation it is a conversion of energy to heat.

2-7C

In order to size the compressor of a new refrigerator, it is desired to determine the rate of heat transfer from the kitchen air into the refrigerated space through the walls, door, and the top and bottom section of the refrigerator. In your analysis, would you treat this as a transient or steady-state heat transfer problem? Also, would you consider the heat transfer to be one-dimensional or multidimensional? Explain.

The problem would be treated as steady-state since temperature would not change with time. This would be considered multidimensional since heat is flowing in multiple directions.

2-8C

In order to determine the size of the heating element of a new oven, it is desired to determine the rate of heat loss through the walls, door, and the top and bottom section of the oven. In your analysis, would you consider this to be a steady or transient heat transfer problem? Also, would you consider this to be one-dimensional or multidimensional? Explain.

This would be considered transient because temperature changes with time. This would be considered multidimensional since heat is flowing in multiple directions.

2-9C

Consider a round potato being baked in an oven. Would you model the heat transfer to the potato as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.

The potato would be three-dimensional since heat is covering all sides of the potato. This would be transient as the potato temperature changes with time. A spherical coordinate system since it is three-dimensional and round shape.

2-10C

Consider an egg being cooked in boiling water in a pan. Would you model the heat transfer to the egg as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.

The egg would be three-dimensional since heat is covering all sides of the egg. This would be transient since the egg temperature changes with time. A spherical coordinate system since it is three-dimensional and round shape.

2-11C

Consider a hot dog being cooked in boiling water in a pan. Would you model the heat transfer to the hot dog as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to solve this problem, and where would you place the origin? Explain.

The hot dog would be two dimensional since the axial direction can be ignored. This would be transient since the hot dog temperature changes with time. A cylindrical coordinate system would be used, and the origin would be along the centerline.

2-12C

Consider the cooking process of roast beef in an oven. Would you consider this to be a steady or transient heat transfer problem? Also, would you consider this to be one-, two-, or three-dimensional? Explain.

This would be transient since the roast beef temperature changes with time. The roast beef would be three-dimensional since the roast beef is heated from all sides.

2-13C

Consider heat loss from a 200-L cylindrical hot water tank in a house to the surrounding medium. Would you consider this to be a steady or transient heat transfer problem? Also, would you consider this heat transfer problem to be one-, two-, or three-dimensional? Explain.

This would be transient heat transfer since the temperature of the water inside the tank would change with time. The hot water tank would be considered two-dimensional as the axial direction can be ignored.

2-14C

Consider a cold canned drink left on a dinner table. Would you model the heat transfer to the drink as one-, two-, or three-dimensional? Would the heat transfer be steady or transient? Also, which coordinate system would you use to analyze this heat transfer problem, and where would you place the origin? Explain.

The cold canned drink would be two-dimensional since the axial direction can be ignored. A cylindrical coordinate system would be used with the origin along the centerline. The heat transfer would be transient since the temperature of the cold canned changes with time.

Heat Conduction Equation

2-20C

Write down the one-dimensional transient heat conduction equation for a plane wall with constant thermal conductivity and heat generation in its simplest form, and indicate what each variable represents.

Constant conductivity:
$$\frac{\partial^2 T}{\partial x^2} + \frac{\dot{e}_{gen}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (2-14)$$

T = temperature

x = spatial coordinate through the wall (rectangular coordinate)

t = time

k = thermal conductivity (constant)

$\dot{e}_{gen}$ = heat generated per unit volume / total rate of heat generation

α = thermal diffusivity, $\alpha = \frac{k}{\rho c}$

2-21C

Write down the one-dimensional transient heat conduction equation for a plane wall with constant thermal conductivity and heat generation in its simplest form, and indicate what each variable represents.

Constant conductivity:
$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\dot{e}_{\text{gen}}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (2-26)$$

T = temperature

r = radial coordinate

t = time

k = thermal conductivity (constant)

$\dot{e}_{\text{gen}}$ = heat generated per unit volume / total rate of heat generation

α = thermal diffusivity, $\alpha = \frac{k}{\rho c}$

Boundary & Initial Conditions; Formulation of Heat Conduction Problems

2-35C

What is a boundary condition? How many boundary conditions do we need to specify for a two-dimensional heat conduction problem?

A boundary condition specifies the thermal conditions on the surfaces of the medium. For a two-dimensional problem 4 boundary conditions must be specified, two in each direction.

2-36C

What is an initial condition? How many initial conditions do we need to specify for a two-dimensional heat conduction problem?

The beginning condition of a process within the medium. Only one initial condition needs to be specified regardless of the number of dimensions.

2-37C

What is a thermal symmetry boundary condition? How is it expressed mathematically?

The temperature in one half is the same as the other. There is no heat flow/transfer across the plates.

$$\frac{\partial T(L/2, t)}{\partial x} = 0 \quad (2-50)$$

2-38C

How is the boundary condition on an insulated surface expressed mathematically?

$$k \frac{\partial T(0, t)}{\partial x} = 0 \quad \text{or} \quad \frac{\partial T(0, t)}{\partial x} = 0 \quad (2-49)$$

2-39C

It is claimed that the temperature profile in a medium must be perpendicular to an insulated surface. Is this a valid claim? Explain.

Yes, an insulated surface is in the normal or perpendicular direction since the slope of the temperature at the surface must be 0.

2-40C

Why do we try to avoid the radiation boundary conditions in heat transfer analysis?

To avoid complications with nonlinearity. This is especially the case when convection is dominant and radiation is minor.

Solution of Steady One-Dimensional Heat Conduction Problems

2-53C

It is stated that the temperature in a plane wall with constant thermal conductivity and no heat generation varies linearly during steady one-dimensional heat conduction. Will this still be the case when the wall loses heat by radiation from its surfaces?

No, since radiation boundary conditions are nonlinear. Therefore, the temperature distribution within the wall is nonlinear.

2-54C

Consider one-dimensional heat conduction through a large plane wall with no heat generation that is perfectly insulated on one side and is subjected to convection and radiation on the other side. It is claimed that under steady conditions, the temperature in a plane wall must be uniform (the same everywhere). Do you agree with this claim? Why?

No. Since there must be a temperature gradient for heat transfer to exist. This would contradict the presence of heat loss since there would be a uniform temperature which wouldn't be correct.

2-55C

Consider a solid cylindrical rod whose side surface is maintained at a constant temperature while the end surfaces are perfectly insulated. The thermal conductivity of the rod material is constant and there is no heat generation. It is claimed that the temperature in the radial direction within the rod will not vary during steady heat conduction. Do you agree with this claim? Why?

Yes, since the temperature is constant and the end surfaces are perfectly insulated. With no heat generation or heat loss the temperature is uniform under steady-state conditions in the radial direction.

2-56C

Consider a solid cylindrical rod whose ends are maintained at constant but different temperatures while the side surface is perfectly insulated. There is no heat generation. It is claimed that the temperature along the axis of the rod varies linearly during steady heat conduction. Do you agree with this claim? Why?

Yes, since there is a temperature difference at both ends and the side of the rod is perfectly insulated with no heat generation, the temperature varies linearly during steady-state conditions.

Heat Generation in a Solid

2-78C

What is heat generation? Give some examples.

A medium which heat is conducted from energy into thermal energy. Some examples include mechanical, electrical, nuclear, or chemical energy.

2-79C

Does heat generation in a solid violate the first law of thermodynamics, which states that energy cannot be created or destroyed? Explain.

No, since heat generation is not created but rather converted from one form of energy into thermal energy.

2-80C

Consider uniform heat generation in a cylinder and a sphere of equal radius made of the same material in the same environment. Which geometry will have a higher temperature at its center? Why?

A sphere will have a higher temperature at its center since it has a smaller surface area relative to its volume making it poor at dissipating heat that is generated.

2-81C

An iron is left unattended and its base temperature rises as a result of resistance heating inside. When will the rate of heat generation inside the iron be equal to the rate of heat loss from the iron?

The rate will be equal when the iron reaches steady-state conditions, when temperature is uniform.

2-82C

Consider the uniform heating of a plate in an environment at a constant temperature. Is it possible for part of the heat generated in the left half of the plate to leave the plate through the right surface? Explain.

Yes, since temperature always goes from high to low no matter where the heat is generated.

Variable Thermal Conductivity

2-104C

Is the thermal conductivity of a medium, in general, constant or does it vary with temperature?

The thermal conductivity of a medium is generally varying with temperature since the material often changes with varying temperatures.

2-105C

When the thermal conductivity of a medium varies linearly with temperature, is the average thermal conductivity always equivalent to the conductivity value at the average temperature?

No, the average temperature is not always equal to the average thermal conductivity since the thermal conductivity can vary with temperature.

2-106C

The temperature of a plane wall during steady one-dimensional heat conduction varies linearly when the thermal conductivity is constant. Is this still the case when the thermal conductivity varies linearly with temperature?

No, when the thermal conductivity varies linearly with temperature the heat conduction equation becomes nonlinear since the distribution of heat within the plane wall is nonlinear.

2-107C

Consider steady one-dimensional heat conduction in a plane wall in which the thermal conductivity varies linearly. The error involved in heat transfer calculations by assuming constant thermal conductivity at the average temperature is (a) none, (b) small, or (c) significant.

(b) small, since the variation of k is small evaluating at average temperature is fine as it gives a good estimate/approximation.

2-108C

Consider steady one-dimensional heat conduction in a plane wall, long cylinder, and sphere with constant thermal conductivity and no heat generation. Will the temperature in any of these mediums vary linearly? Explain.

Plane Wall: Yes, since the general solution to the differential equation is a straight line.

Long Cylinder: No, since the general solution to the differential equation is logarithmic

Sphere: No, since the general solution to the differential equation is an inverse radial ($1/r$).

Review of Differential Equations

2-121C

Why do we often utilize simplifying assumptions when we derive differential equations?

Simplifying assumptions reduces the complexity of the differential equation. This allows for the equation to be solved with system conditions in a less complex way.

2-122C

What is a variable? How do you distinguish a dependent variable from an independent one in a problem?

A variable is a quantity that can change. A dependent variable changes because of the independent variable.

2-123C

Can a differential equation involve more than one independent variable? Can it involve more than one dependent variable? Give examples.

Yes, a differential equation can involve more than one independent variable and dependent variable. An example for independent is temperature $T(x, y, z, t)$ which depends on space and time. An example for dependent is pressure and velocity components in fluid flow in mass conservation & Navier-Stokes.

2-124C

What is the geometrical interpretation of a derivative? What is the difference between partial derivatives and ordinary derivatives?

The geometrical interpretation is the slope of a tangent line at a point on a curve. An ordinary derivative depends on only one independent variable, while a partial derivative depends on multiple independent variables.

2-125C

What is the difference between the degree and the order of a derivative?

An order of a derivative is the number of times the dependent variable is differentiated (y'''), while a degree of a derivative is the power of highest-order derivative $(y')^3$.

2-126C

Consider a function $f(x, y)$ and its partial derivative $\partial f / \partial x$. Under what conditions will this partial derivative be equal to the ordinary derivative df/dx ?

The partial derivative is only equal to the ordinary derivative when the function f only depends on x .

2-127C

Consider a function $f(x)$ and its derivative df/dx . Does this derivative have to be a function of x ?

No the derivative does not have to be a function of x , since the derivative only shows how fast something is changing which can be constant or changing.

2-128C

How is integration related to derivation?

Integration is the inverse of derivation.

2-129C

What is the difference between an algebraic equation and a differential equation?

An algebraic equation relates variables and fixed numbers, while differential equations models rate of change for a function.

2-130C

What is the difference between an ordinary differential equation and a partial differential equation?

An ordinary derivative depends on only one independent variable, while a partial derivative depends on multiple independent variables.

2-131C

How is the order of a differential equation determined?

The order is determined by the highest derivative in an equation.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial T}{\partial x} + T = 0, \text{ Order} = 2nd \text{ Order}$$

2-132C

How do you distinguish a linear differential equation from a nonlinear one?

A linear differential equation is when the derivative is to the first power. A nonlinear differential equation is when the derivative is not to the first power.

2-133C

How do you recognize a linear homogeneous differential equation? Give an example and explain why it is linear and homogeneous.

A linear homogeneous differential equation is an equation where all terms are dependent or its derivative.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial T}{\partial x} + T = 0$$

This is linear and homogeneous because all dependent variables (T) are to the first power and is homogeneous since it only contains derivatives of the dependent variables and the variable (T) itself.

2-134C

How do differential equations with constant coefficients differ from those with variable coefficients? Give an example for each type.

When the coefficients of a differential equation are constant, the dependent variable and derivative do not depend on an independent variable.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial T}{\partial x} + T = 0$$

When coefficients of a differential equation are variables, the coefficient depends on an independent variable.

$$x \frac{\partial^2 T}{\partial x^2} + \frac{\partial T}{\partial x} + T = 0$$

2-135C

What kind of differential equations can be solved by direct integration?

A differential equation that can be solved by direct integration is a first order ordinary differential equation.

2-136C

Consider a third-order linear and homogeneous differential equation. How many arbitrary constants will its general solution involve?

The general solution for a third-order linear & homogeneous differential equation is 3 arbitrary constants.